Physics 9702 Paper 4 · Topic 24

Medical physics

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/42 October/November 2020, Question 7 (6 marks) page 2

Examiner report: (a) A very common misconception held by candidates was that the electrons arriving at the target anode had a range of different kinetic energies. Only the strongest candidates realised that the spikes are due to excitation and de-excitation of electrons in the target atoms.

2 9702/42 May/June 2023, Question 10 (9 marks) page 3

Examiner report: (a) (i) Most candidates correctly identified the charged particle as an electron, although a significant minority of weaker candidates confused the whole process of X-ray production with positron emission tomography, and gave the positron as the answer here. (ii) The process of X-ray production by electron bombardment of a metal target was generally well known, although many candidates found it difficult to articulate that the energy of the emitted X-ray photons comes from the kinetic energy of the electrons.

3 9702/42 February/March 2021, Question 11 (10 marks) page 4

Examiner report: (a) (i) There was often confusion with the explanation for the intensity spikes in the emission spectrum, and many candidates gave explanations of the spikes rather than the continuous spectrum. Of those who answered the question as it was asked, a common misconception was that the electrons arrive at the target anode with different speeds / kinetic energies (despite the fact that they have all been accelerated through the same p.d.).

4 9702/41 May/June 2022, Question 9 (9 marks) page 6

Examiner report: (ii) Combining transmitted intensities through more than one medium proved to be challenging.

5 9702/42 October/November 2025, Question 10 (9 marks) page 8

Examiner report: (ii) Many candidates found this a challenging question and found it difficult to set up the starting equation. Some made the correct start but then had difficulty with the arithmetic.

6 9702/44 October/November 2025, Question 8 (15 marks) page 10

Examiner report: (a) Candidates found it difficult to give a complete and correct meaning of a tracer. Absorption into the tissue ‘being studied’ or equivalent wording was rarely stated.

7 9702/42 February/March 2020, Question 4 (8 marks) page 12

Examiner report: (a) (i) Only a minority of candidates were able to explain why ultrasound in scanning is emitted in pulses. The majority of candidates explained how the image is formed rather than how the reflected pulse is detected.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

7 Electrons in a beam are travelling at high speed in a vacuum. The electrons are incident on a metal target, causing X-ray radiation to be emitted.

The variation with wavelength λ of the intensity I of the emitted X-ray radiation is shown in Fig. 7.1.

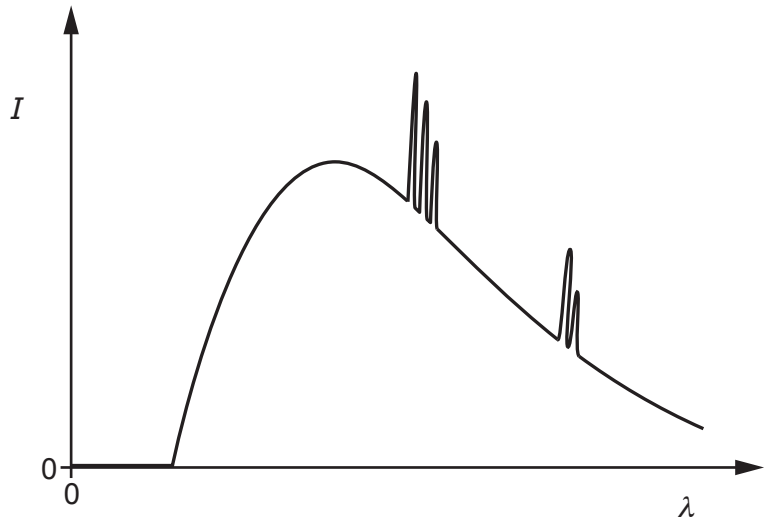


Fig. 7.1

Explain why:

(a) there is a continuous distribution of wavelengths

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..... [3]

(b) at certain wavelengths, there are narrow peaks of increased intensity.

.....

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..... [3]

[Total: 6]

10 **(a)** X-rays for use in medical diagnosis are produced in an X-ray tube. In the X-ray tube, charged particles are accelerated towards a metal target by an applied potential difference (p.d.).

(i) State the name of the charged particles that are accelerated by the applied p.d.

..... [1]

(ii) Explain how X-rays are produced at the metal target.

.....

.....

.....

..... [2]

(iii) Calculate the minimum wavelength of X-rays produced when the applied p.d. is 5.80 kV.

wavelength = ... m [3]

(b) X-rays pass through a medium that has an attenuation coefficient of 1.4 cm^{-1}.

Calculate the percentage of the X-ray energy that is **absorbed** by a 2.8 cm thickness of this medium.

percentage absorbed = ... % [3]

[Total: 9]

11 **(a)** Electrons are accelerated through a potential difference of 15 kV. The electrons collide with a metal target and a spectrum of X-rays is produced.

(i) Explain why a continuous spectrum of energies of X-ray photons is produced.

.....

.....

.....

.....

..... [3]

(ii) Calculate the wavelength of the highest energy X-ray photon produced.

wavelength = ... m [3]

(b) A beam of X-rays has an initial intensity I_o. The beam is directed into some body tissue. After passing through a thickness x of tissue the intensity is I. The graph in Fig. 11.1 shows the variation with x of ln (I/I_o).

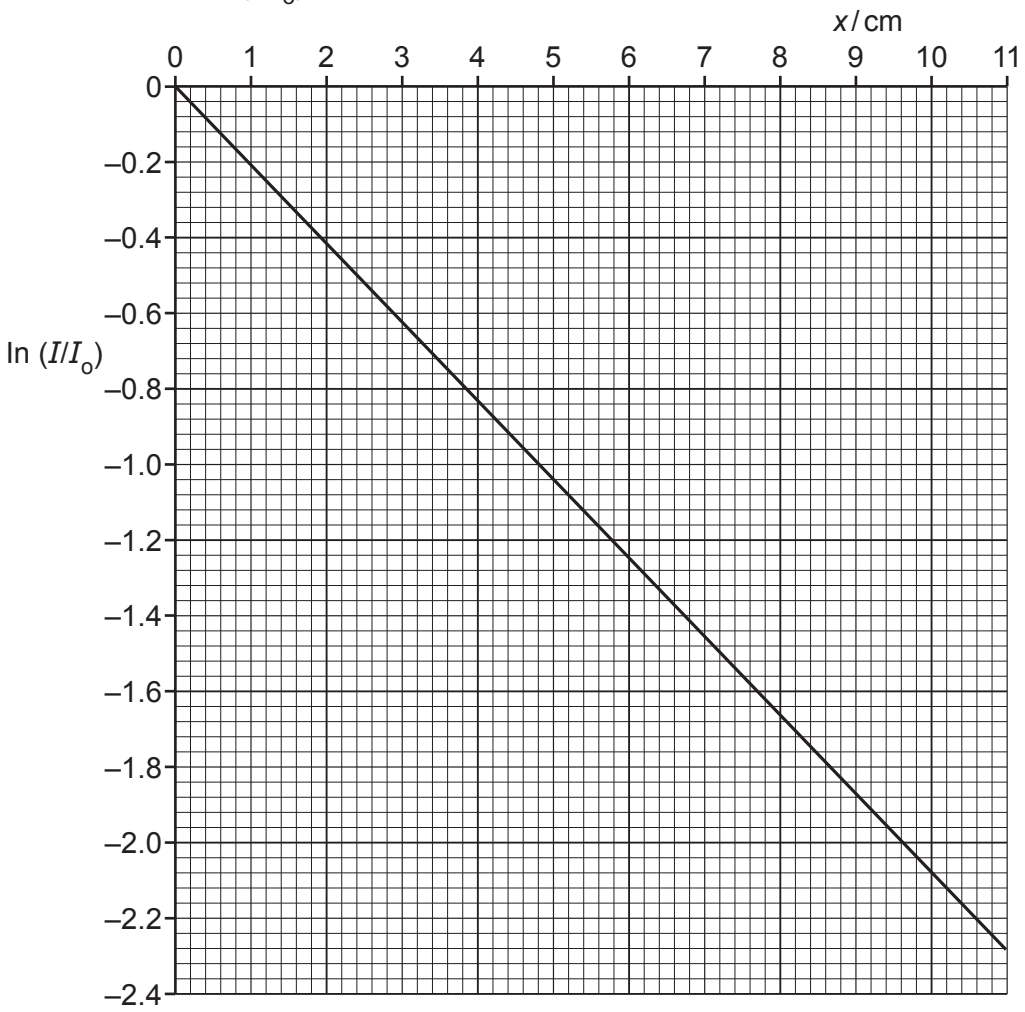


Fig. 11.1

(i) Determine the linear attenuation (absorption) coefficient μ for this beam of X-rays in the tissue.

μ = ... cm^{-1} [2]

(ii) Determine the thickness of tissue that the X-ray beam must pass through so that the intensity of the beam is reduced to 5.0% of its initial value.

thickness = ... cm [2]

[Total: 10]

9 **(a)** **(i)** Explain how X-rays are produced for use in medical diagnosis.

.....

.....

.....

..... [3]

(ii) State why X-ray images are taken of multiple sections of the body during computed tomography (CT) scanning.

.....

..... [1]

(b) An X-ray image is taken of the structure shown in Fig. 9.1.

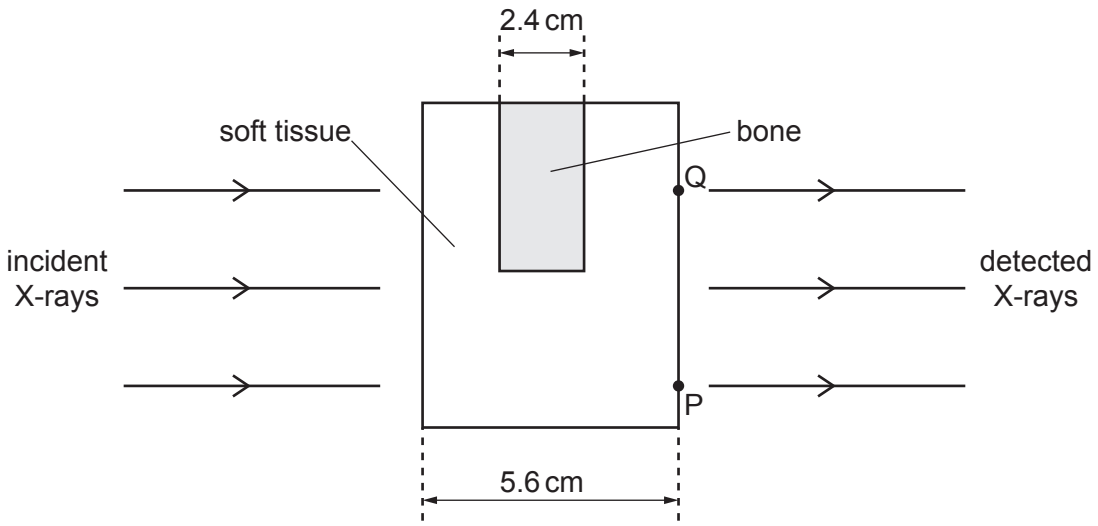


Fig. 9.1

The linear attenuation coefficient of bone is $3.4\,\text{cm}^{-1}$.
The linear attenuation coefficient of soft tissue is $0.89\,\text{cm}^{-1}$.

The incident X-rays are parallel and have a uniform intensity I_0 across the structure.

Determine, in terms of I_0, the intensity of the detected X-rays from:

(i) point P

detected intensity = ... I_0 [2]

(ii) point Q.

detected intensity = ... I_0 [2]

(c) Explain, with reference to your answers in **(b)**, whether the X-ray image of the structure in Fig. 9.1 has good contrast.

.....

.....

..... [1]

[Total: 9]

10 **(a)** State what is meant by contrast in an X-ray image.

.....

..... [1]

(b) X-rays of intensity I_0 are incident normally on a structure, as shown in Fig. 10.1.

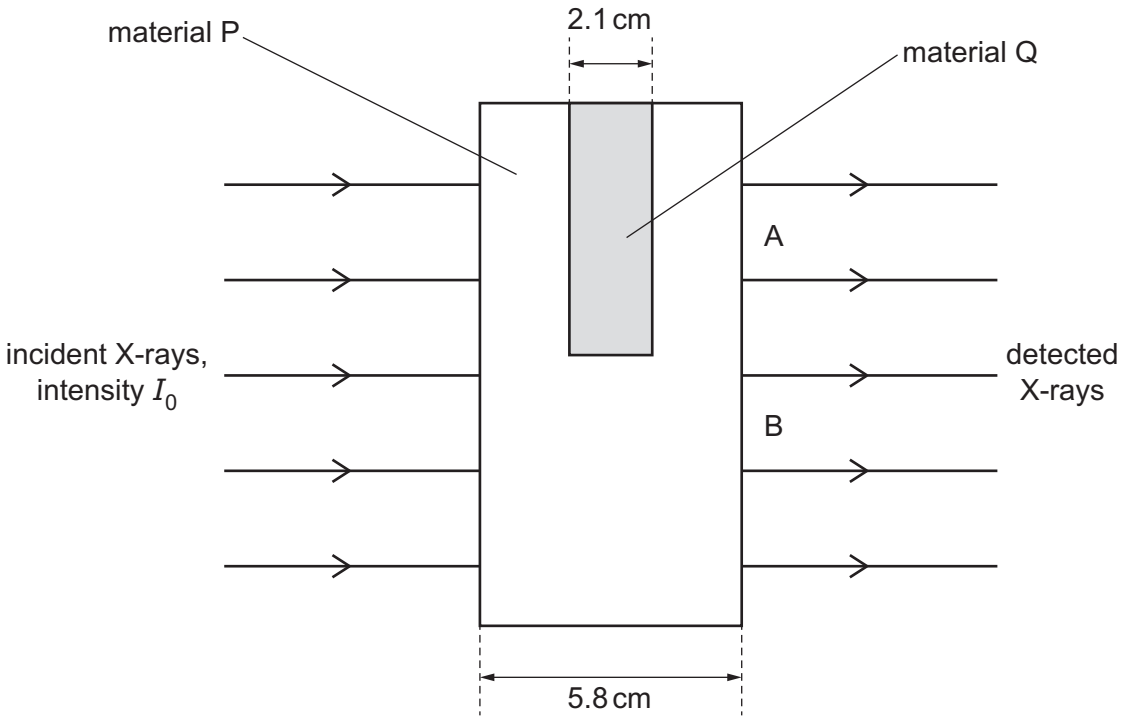


Fig. 10.1

Material P has a linear attenuation coefficient of 0.35 cm^{-1}.
The X-rays emerging from the structure in region A have an intensity of $0.053I_0$.

(i) Show that the intensity of the X-rays emerging in region B is $0.13I_0$.

[1]

(ii) Determine the linear attenuation coefficient μ of material Q.

μ = ... cm^{-1} [3]

(iii) Use the information in **(b)(i)** to suggest why the X-rays emerging from the structure form an image that has poor contrast.

.....

.....

..... [1]

(c) Explain how X-rays are used in computed tomography (CT) scanning to produce a three-dimensional image of an internal structure.

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.....

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..... [3]

[Total: 9]

8 Oxygen-15 $\left({}^{15}_{8}\mathrm{O}\right)$ is radioactive and has a half-life of 2.04 minutes.

The decay of oxygen-15 produces positrons. For this reason, oxygen-15 is sometimes used as a tracer in positron emission tomography (PET scanning).

(a) State what is meant by a tracer.

.....

.....

..... [2]

(b) The equation for the decay of oxygen-15 is

$$ {}^{15}_{8}\mathrm{O} \rightarrow {}^{P}_{Q}\mathrm{X} + {}^{R}_{S}\beta^{+} + \mathrm{Z} $$

where X is the nucleus formed during the decay and Z is another particle.

(i) State the values of the integers *P*, *Q*, *R* and *S*.

P = ... *R* = ...

Q = ... *S* = ...

[2]

(ii) State the name of particle Z.

..... [1]

(c) **(i)** Define the activity of a sample.

.....

..... [1]

(ii) Calculate the decay constant of oxygen-15. Give a unit with your answer.

decay constant = ... unit ..... [2]

(iii) Determine the rate at which positrons are produced in a sample of oxygen-15 that has a mass of 2.85×10^{-6} kg.

rate = ... s^{-1} [4]

(d) The particles that are emitted from the body and detected outside it during PET scanning are not positrons but another type of particle.

(i) State the name of the particles that are detected.

..... [1]

(ii) Explain how these particles are formed inside the body.

.....

.....

.....

..... [2]

[Total: 15]

4 **(a)** **(i)** Explain why ultrasound used in medical diagnosis is emitted in pulses.

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..... [2]

(ii) Explain the principles of the **detection** of ultrasound waves used in medical diagnosis.

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..... [3]

(b) The specific acoustic impedances Z of some media are given in Table 4.1.

Table 4.1

media	Z / kg m^{-2} s^{-1}
air	4.3×10^2
gel	1.5×10^6
soft tissue	1.6×10^6

(i) The specific acoustic impedances of two media are Z_1 and Z_2. The intensity reflection coefficient α for the boundary of these two media is given by:

$$\alpha = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}.$$

Calculate, to three significant figures, the fraction of the ultrasound intensity that is reflected at a boundary between air and soft tissue.

α = ... [1]

(ii) Use your value in **(b)(i)** to explain why gel is applied to the surface of the skin during an ultrasound scan.

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.....

.....

..... [2]

[Total: 8]

Question	Answer	Marks
7(a)	X-ray photon produced when electron is decelerated	**B1**
	larger acceleration results in larger photon energy	**B1**
	continuous range of accelerations so continuous spectrum of wavelengths/frequencies	**B1**
7(b)	electron in (inner shell of) target atom is excited (on collision)	**B1**
	electron de-excites causing emission of a photon	**B1**
	discrete energy levels so discrete photon wavelengths	**B1**

Question	Answer	Marks
10(a)(i)	electrons	**B1**
10(a)(ii)	electrons are decelerated / stopped on impact with the target	**B1**
	(kinetic) energy lost by electrons emitted as (X-ray) photons	**B1**
10(a)(iii)	$eV = hc / \lambda$	**C1**
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^{8}) / (1.60 \times 10^{-19} \times 5800)$	**C1**
	$= 2.14 \times 10^{-10}$ m	**A1**
10(b)	$I = I_0 \exp(-\mu x)$	**C1**
	$I_T / I_0 = \exp(-(1.4 \times 2.8))$ $= 0.020$	**C1**
	% absorbed $= (1.000 - 0.0198) \times 100$ $= 98\%$	**A1**

Question	Answer	Marks
11(a)(i)	electrons decelerate (on hitting target) so X-ray photons produced	**B1**
	range of decelerations	**B1**
	photon energy depends on (magnitude of) deceleration	**B1**
11(a)(ii)	$eV = \frac{hc}{\lambda}$	**C1**
	$\lambda = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{1.6\times10^{-19}\times15000}$	**C1**
	$= 8.3\times10^{-11} m$	**A1**
	or $E = hf$ and $c = f\lambda$ and electron energy = eV *or* $E = hc / \lambda$ and electron energy = eV electron energy = $1.6 \times 10^{-19} \times 15000$ $= 2.4 \times 10^{-15}$	**(C1)**
	$\lambda = \frac{6.63\times10^{-34}\times3.0\times10^{8}}{2.4\times10^{-15}}$	**(C1)**
	$\lambda = 8.3\times10^{-11} m$	**(A1)**
11(b)(i)	μ = – gradient *or* ln $(I / I_o) = -\mu x$	**C1**
	(e.g. 2.08 / 10.0) = **0.21** cm^{-1}	**A1**

Question	Answer	Marks
11(b)(ii)	$\ln(0.05) = -\mu x$	**C1**
	$x = \dfrac{\ln 0.05}{-\mu}$ $e.g.\ x = 14\,cm$	**A1**

Question	Answer	Marks
9(a)(i)	electrons are accelerated (by an applied p.d.)	**B1**
	electrons hit target	**B1**
	X-rays produced when electrons decelerate	**B1**
9(a)(ii)	images of the multiple sections are combined to create a 3-D image	**B1**
9(b)(i)	$I = I_0 \exp(-\mu x)$	**C1**
	$= I_0 \exp(-0.89 \times 5.6)$ $= 0.0068\ I_0$	**A1**
9(b)(ii)	$I = I_0 \exp(-2.4 \times 3.4) \times \exp(-0.89 \times 3.2)$	**C1**
	$= 1.7 \times 10^{-5}\ I_0$	**A1**
9(c)	comparison of intensities or values in **(b)** leading to conclusion consistent with these values	**B1**

Question	Answer	Marks
10(a)	<u>difference</u> in degrees of blackening	**B1**
10(b)(i)	$I = I_0 \exp(-\mu x)$ $= I_0 \exp(-5.8 \times 0.35) = 0.13\ I_0$	**A1**
10(b)(ii)	use of exp {–(0.35 × 3.7)} factor	**C1**
	$0.053 I_0 = I_0 \exp\{-[(0.35 \times 3.7) + 2.1\mu]\}$	**C1**
	$\mu = 0.78\ \text{cm}^{-1}$	**A1**
10(b)(iii)	factor of only 2.5 between the (detected) <u>intensities</u> (so not good contrast)	**B1**
10(c)	(structure) scanned in (thin) sections	**B1**
	(many) scans (of each section) taken from different angles	**B1**
	scanning repeated for all sections and (data) compiled (to form 3D image)	**B1**

Question	Answer	Marks
8(a)	(radioactive) substance introduced into the body	**B1**
	substance absorbed by the tissues being studied	**B1**
8(b)(i)	$P = 15$ **and** $R = 0$	**A1**
	$Q = 7$ **and** $S = (+)1$	**A1**
8(b)(ii)	(electron) neutrino	**B1**
8(c)(i)	number of nuclear disintegrations per unit time	**B1**
8(c)(ii)	decay constant $= \ln 2 / (2.04 \times 60)$	**C1**
	$= 5.66 \times 10^{-3}\ s^{-1}$	**A1**
8(c)(iii)	$N = (2.85 \times 10^{-6}) / (15 \times 1.66 \times 10^{-27})$	**C1**
	$A = \lambda N$	**C1**
	rate $= (5.66 \times 10^{-3}) \times (2.85 \times 10^{-6}) / (15 \times 1.66 \times 10^{-27})$	**C1**
	$= 6.48 \times 10^{17}\ s^{-1}$	**A1**
8(d)(i)	gamma photons	**B1**
8(d)(ii)	positron collides with electron (in body)	**B1**
	annihilation results in their masses becoming photon energy	**B1**

Question	Answer	Marks
4(a)(i)	Any 2 from: • allows the reflected signal to be distinguished from the emitted signal • detection occurs in the time between emitted pulses • (reflection of ultrasound) detected by same probe / transducer / crystal • cannot emit and detect at same time (hence pulses)	**B2**
4(a)(ii)	piezo-electric crystal	**B1**
	ultrasound makes crystal vibrate / resonate	**B1**
	vibration produces (alternating) e.m.f. / p.d. across crystal	**B1**
4(b)(i)	= $(1.6 \times 10^6 - 4.3 \times 10^2)^2 / (1.6 \times 10^6 + 4.3 \times 10^2)^2$ = **0.999**	**B1**
4(b)(ii)	without the gel most of the ultrasound is reflected	**B1**
	Z values more similar / α reduces so less (ultrasound) is reflected / more (ultrasound) is transmitted	**B1**